



Management of rare coagulation disorders in 2018

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ABSTRACT

Rare bleeding disorders (RBDs) comprise inherited deficiencies of factors I (fibrinogen), II (prothrombin), V, VII, X, XI, and XIII as well as combined factor V + VIII and vitamin K-dependent factors. They represent 3–5% of all congenital bleeding disorders and are usually transmitted as autosomal recessive traits. These disorders often manifest during childhood and have varied clinical presentations from mucocutaneous bleeding to life-threatening symptoms such as central nervous system and gastrointestinal bleeding. Bleeding manifestations generally vary within the same RBD and may also vary from 1 RBD to the other. Laboratory diagnosis is based on coagulation screening tests and specific factor assays, with molecular techniques providing diagnostic accuracy and enabling prenatal counseling. The approach to treatment of bleeding episodes and invasive procedures needs to be individualized and depends on the severity, frequency and procedure-related risk of bleeding. The first line of treatment of RBDs is replacement of the deficient factor, using specific plasma-derived or recombinant products and using fresh frozen plasma or cryoprecipitate when specific products are not available or in resource-limited countries. Prophylaxis may be considered in individuals with recurrent serious bleeding and especially after life-threatening bleeding episodes. Novel no-replacement strategies promoting hemostasis by through different mechanisms need to be studied in RBDs as alternative therapeutic options.

1. Introduction

Rare bleeding disorders (RBDs) consist of inherited quantitative or qualitative deficiencies of coagulation factors such as fibrinogen or Factor (F) I, prothrombin or FII, FV, FVII, FX, FXI, FXIII and combined factor deficiencies such as FV + FVIII or FII, FVII, FIX, and FX (vitamin K-dependent coagulation factors deficiency [VKCFD]) [1]. The identification of these disorders is challenged by their rarity, clinical heterogeneity, inadequacy of available laboratory assays and a frequent lack of family history to suggest an inherited bleeding disorder. The establishment of registries has vastly improved our understanding of these disorders but gaps still exist pertaining to the natural history and management of RBDs (from the lack of recognition in the community) even when followed at centers which specialize in the diagnosis and management of these disorders [2]. This review summarizes the epidemiology, clinical presentation, and diagnosis, with an additional focus on women's health issues. We also describe the currently available treatments for RBDs and discuss target levels for prophylaxis vs on demand therapy and emerging alternative non-factor replacement therapies. Inhibitor development, a worrisome complication in rare bleeding disorders which is less well characterized and poorly

understood as compared to hemophilia, is also addressed. Detailed discussion on the diagnosis and management of congenital FVII and FXIII deficiencies will be published in Volume 2 of this theme edition on Hereditary Bleeding Disorders for Transfusion and Apheresis Science in December.

2. Epidemiology and clinical presentation

Our knowledge of the worldwide distribution of RBDs has been enhanced by data collected through two global surveys conducted by the World Federation of Haemophilia (WFH) (<http://wfh.org>) and the Rare Bleeding Disorders Database (RBDD) (www.rbdd.org). The worldwide prevalence of RBDs has been estimated to be between 3–5% of all inherited coagulation disorders with a recent report of 8% derived from the WFH survey in 2016 [3–5]. Increased prevalence of RBDs is particularly noted in areas with a high rate of consanguinity and endogamous marriages, such as the Middle-East, Arabian Peninsula, North Africa, and India [6–14] and historically in outports of Newfoundland and Labrador, the most Eastern provinces of Canada [15]. The prevalence of 1 symptomatic person per 500, 000 population (homozygous or compound heterozygous) for FVII deficiency makes it the most

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common RBD, constituting 28 to 36% of RBDs in published series [3–5,16–20]. This is closely followed by FXI deficiency which accounts for 23 to 32% of all RBDs and less prevalent are deficiencies of FI, FV and FX (7–10% each), FXIII (~6%) [3,4,17,19], and FV + FVIII deficiency (~2–3%). The rarest of the RBDs is FII deficiency with a prevalence of ~1 per 2 million (2%) [4,17,19]. There are less than 50 families worldwide with VKCFD that have been reported in the literature [21,22].

There are no inherent race- or ethnicity based propensities for specific RBDs, with the exception of FXI deficiency, which is prevalent among Ashkenazi Jews; 1 out of every 450 individuals is affected and 8% cases are heterozygous, accounting for a higher incidence of severe disease [23–25]. In addition, FII deficiency was found to disproportionately affect Latinos in a survey of fifty-eight North American hemophilia treatment centers [5].

2.1. Inheritance

Most RBDs are inherited as autosomal recessive (AR); however, heterozygotes may have varying degrees of corresponding factor deficiency that render an unpredictable propensity for bleeding. Some cases of FXI deficiency and dysfibrinogenemia may be autosomal dominant conferring an autosomal variable pattern of expression [4]. Combined FV + FVIII deficiency and VKCFD are caused by mutations in genes outside those encoding the coagulation factors themselves and are true AR disorders in that heterozygosity confers normal factor levels making them asymptomatic.

2.2. Clinical Manifestations of RBDs

Clinically, RBDs are characterized by a diverse range of bleeding symptoms, from none or minimal to life-threatening events such as central nervous system (CNS) and musculoskeletal bleeding [17,26,27]. There is significant variation in bleeding patterns between individual RBDs and in individuals with the same type of RBD. The association between clinical bleeding severity of RBDs and residual clotting factor activity levels also varies markedly, which was explored as part of data collected on 489 patients through the European Network of the Rare Bleeding Disorders (EN-RBD) [27]. This data revealed a strong association of clinical bleeding symptoms with FI, FII, FX, and FXIII activity levels, a weak correlation in phenotypically heterogeneous FV and FVII deficiencies and no association with FXI activity [27]. This implies that each RBD has a dissimilar level to ensure absence of clinical symptoms or those that result in major spontaneous bleeding, (unlike hemophilia A or B) and therefore, a single classification for all types of RBDs may not be appropriate [17]. Based on data from EN-RBD, it was proposed that bleeding episodes be classified, based on location, spontaneity of bleeding as well as potential clinical impact, as severe: associated with risk of spontaneous major bleeding, moderate: with spontaneous mild bleeding events, and mild: with bleeding occurring after trauma, drug ingestion and being asymptomatic [17,27]. This classification has some practical utility for most of the RBDs, except FXI deficiency. There is another severity classification proposed for FVII which is based on symptoms rather than activity levels which might be considered for those RBDs that weakly correlate phenotypically with coagulant activity levels and is further discussed in the chapter on FVII deficiency [20].

The optimal assessment of the presence and severity of bleeding symptoms is important in assessing the probability of having a RBD in subjects referred for evaluation of a bleeding disorder. A new bleeding assessment tool (BAT) has been specifically designed for RBDs by Palla et al. from data derived from the EN-RBD database to avoid subjective interpretation of symptoms by patients especially in mild cases [28]. The authors reported that a bleeding score (BS) of 1.5 using the BAT was able to discriminate between patients with and without RBD with a sensitivity of ~ 67% and specificity of ~74% and a negative

correlation was observed between BS and factor activity level [28]. However, further evaluation of its validity and reliability in RBDs is needed.

The most common bleeding symptom across all ages and genotypes of RBDs is mucocutaneous bleeding [4,5,17]. Other frequent occurrences of excessive bleeding are after circumcision or heel stick puncture in the neonatal period, post-surgical including dental procedures, and menstrual or pregnancy related, although the historical absence of bleeding after these challenges does not necessarily exclude a RBD [26,29,30].

In general, FXI deficiencies is rarely associated with serious bleeding manifestations [4,25,27,31,32]. Spontaneous bleeding is uncommon in FXI even at moderately reduced activity levels (< 15–20%) but may present after trauma, surgery, or post-partum especially in tissues with high fibrinolytic activity such as nasopharynx and urogenital tract [2]. Umbilical cord and CNS bleeding are prevalent with severe FI, FII, FX and FXIII deficiencies and has been reported in a few patients with severe FVII deficiency [4,17,20,26,27,33]. Gastrointestinal (GI) bleeding is a common manifestation with severe FX deficiency but has been reported less frequently with all other RBDs except FV + FVIII deficiency [4,27]. Soft tissue and joint bleeding have been reported with all RBDs but are relatively more common in severe FI, FII, FX and FXIII deficiencies [4,26,27,34,35]. Other phenotypic manifestations might be associated with certain RBDs, for instance, FX deficiency has been associated with various non-coagulation-related abnormalities, including thrombocytopenia-absent radius syndrome, mitral valve prolapse, hypertrophic cardiomyopathy, and hypercholesterolemia [36]. With FI deficiency (afib- and dysfibrinogenemia), arterial and venous thromboembolic complications, bone cysts, and splenic infarcts have been reported to occur [27]. There have been cases of myocardial infarction and venous thrombosis reported in patients with cardiovascular risk factors and after FXI concentrate infusion [37–39].

Combined FV + FVIII deficiency is associated with a mild bleeding phenotype, without any apparent cumulative effect of the combined deficiencies [40]. The homozygous phenotype is usually characterized by FV and FVIII factor activity levels in the range of 4–30% and have corresponding antigen levels [40]. The most commonly reported bleeding symptoms are epistaxis and menorrhagia, post-surgical and post-partum bleeding whereas infrequently noted are spontaneous soft tissue and joint bleeds [40]. In contrast, those affected with VKCFD often present in infancy or in early childhood with CNS or umbilical stump bleeding [21,41,42]. Skeletal abnormalities have also been reported, possibly due to defective γ -carboxylation of bone matrix proteins [21].

Given the heterogeneity of the bleeding phenotype in these various RBDs the role of other factors attributing to bleeding phenotype needs to be explored, such as the fibrinolytic potential of the site of injury, activation of the thrombin-activatable fibrinolysis inhibitor with FXI deficiency, possible coexistence of other hemostatic defects such as inherited thrombophilia, additional source of functional FV from platelets, and reduced levels of tissue factor pathway inhibitor associated with FV deficiency.

Bleeding in Women with RBDs

Heavy menstrual bleeding is the most common bleeding symptom reported in ~50% of women with RBDs irrespective of the type of bleeding disorder [43,44]. Affected women can experience chronic iron deficiency anemia or pain, with significantly reduced quality of life due to limitation in social activities, work and education [45]. Women with RBDs are also at a higher risk for developing gynecological conditions associated with bleeding such as hemorrhagic ovarian cysts, endometriosis or structural lesions (eg, leiomyomata) which may lead to significant bleeding [43,44]. Women with RBDs may experience obstetric complications as well, including postpartum hemorrhage [46] and recurrent miscarriages due to the roles that factors, FXIII and FI, play in placental implantation and pregnancy maintenance [34,46–48]. Paradoxically, thrombotic episodes have been reported in patients with

FI and FVII deficiencies in the post-partum and post-surgical settings [48].

2.3. Inhibitor development

Although there is a better understanding of inhibitor development in hemophilia A and B, there is little knowledge pertaining to inhibitors in persons with RBDs given their rarity. The management of the rare occurrence of inhibitor development in RBDs becomes even more challenging for providers with scant available data. Alloantibodies complicating replacement for FV, VII, XI, and FXIII deficiencies are more commonly encountered. These antibodies are of the IgG subclass but have not been well characterized. There have only been anecdotal reports in the literature which describe their development after replacement with plasma and plasma-derived concentrates. Details regarding clinical characteristics and management of inhibitors in these factor deficiencies has been provided in Table 4.

3. Diagnostic evaluation and considerations

3.1. Laboratory diagnosis

The laboratory evaluation for diagnosis of RBDs requires routine coagulation studies (prothrombin time [PT], activated partial thromboplastin time [APTT] which if prolonged can be used to narrow down the possible disorders (see Fig. 1). It is important to note that PT and APTT may not be sensitive enough to pick up FI and FII deficiencies. Another screening assay is the thrombin time which is required to detect fibrinogen deficiency. The factor deficiencies can be quantitative or qualitative, based on the concordance/discordance of factor antigen and activity levels. Specifically in the case of FI and FII deficiencies, antigen levels are necessary to detect a dysfunctional state as normal levels for these two factors can increase the risk of thrombosis, particularly if other prothrombotic risk factors co-exist [17,49].

There are several important considerations in the laboratory evaluation for RBDs, particularly in young children. First, reference values for full-term and preterm infants in the first 6 months of life (and beyond, in some cases) differ from those in adults, particularly for APTT and all coagulation factor levels except for FVIII [50,51]. Because the levels of many factors increase over the first month of life, milder or equivocal deficiencies may need to be confirmed once a child is older or by screening both parents. Another factor to be taken into consideration includes a difficult venipuncture leading to contamination by tissue factor resulting in erroneous results.

Lack of standardization of the available factor assays generates mismatch between levels measured by different assays. This creates uncertainty concerning the criteria by which these RBDs can be classified. Certain recommendations for measuring specific factor assays have been proposed by the Rare Bleeding Disorders Working Group [27]. For measuring plasma level of FI, it is recommended that the

Clauss method be the reference method as it can be performed on all instruments as compared to the PT-derived method. Similarly, the choice of thromboplastin reagent used in FVII activity assays is crucial for diagnostic accuracy. It is recommended to use recombinant thromboplastins structurally identical to human tissue factor [52]. Allo-antibodies to FV, FVII, FXI, and FXIII can be measured using the Bethesda assay [53].

Global measurement of thrombin generation (TG) instead of isolated factor activity can help in better defining hemorrhagic risk by providing an overall view of hemostasis. These assays enable assessment of clot kinetics, fibrinolysis and individual hemostatic capacity which may correlate better with symptom severity. Various techniques of measuring TG have been studied in predicting bleeding risk in patients with RBDs and to determine treatment effectiveness [54–60]. Rotating thromboelastogram (ROTEM) technique was utilized for monitoring therapy with bypassing agents in patients with severe FXI deficiency [55,56]. There is a continued effort towards the standardization of these methods which can then be utilized to monitor treatment in various bleeding situations in RBDs, particularly those associated with thrombotic risk such as FI and FXI deficiencies [32,34,61–63].

3.2. Molecular diagnosis

The molecular diagnosis of RBDs is based on identification of the causative mutation in the genes encoding the corresponding coagulation factors. There are two exceptions to this: 1) combined FV + FVIII deficiency which stems from a common molecular defect affecting intracellular processing of both proteins (MCFD2 and LMAN1) [64]; 2) VKCFD which has 2 subtypes based on the mutated enzyme involved in vitamin K metabolism, namely GGX and VKORC1 [21,41,42]. Amongst the different RBDs, missense mutations are the most prevalent, being present in 52–57% of patients followed by small insertions/deletions in 20–30% mostly found in FI and FV deficiencies [64]. Splicing and nonsense mutations are not very common, each accounting for a prevalence of < 15% of patients [64].

Prenatal diagnosis can be offered in most inherited RBDs by obtaining fetal DNA via amniocentesis or chorionic villus sampling and testing for known familial mutations [65]. It can also be analyzed to compare linkage and sequence against the proband and parents. Early diagnosis can allow decision making regarding continuation of pregnancy or preparing with appropriate intervention and therapy, as indicated. Moreover, in vitro fertilization can enable selection of healthy embryos by using pre implantation genetic determination [65–67].

Detailed gene mutation analysis by collectively obtaining data from various national, international databases is important to get information on common and recurrent mutations for each disorder in each specific region of the world. This would allow for these genetic variants to be used as markers for quick and convenient genotyping. For example with the fibrinogen disorders null mutations of the FGA gene

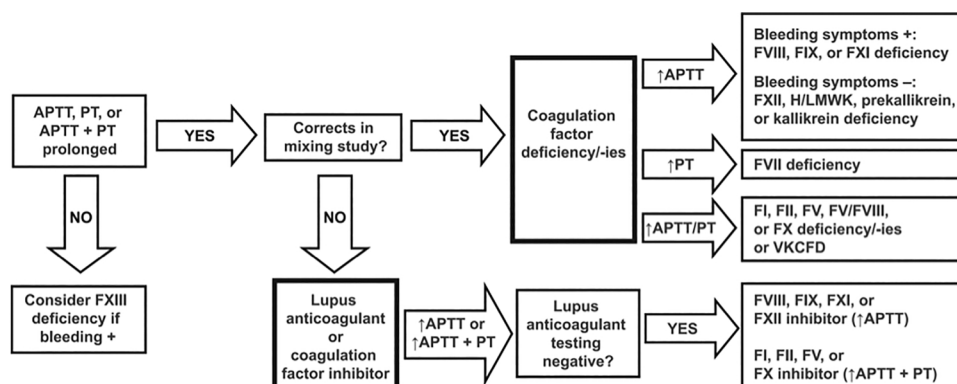


Fig. 1. Proposed laboratory diagnostic algorithm for evaluating coagulation profile abnormalities. A mixing study should be performed to differentiate a coagulation factor deficiency from an inhibitor. The screening assays help to narrow down the possible factor involved for which a specific factor assay can be performed. Note: PT and APTT are normal in FXIII deficiency. H/LMWK, high-/low-molecular-weight-kininogen. Reprinted with permission from Pediatrics 2013, 132 (5), 882–892.

Table 1

Factor Replacement Products Available in the United States and Internationally for Treatment or Prevention of Bleeding in RBDs.

Replacement Product	Content	Viral Inactivation	Therapeutic Option for	
			First-line	Second-line
Single-factor concentrates	pd-fibrinogen	Yes	Fibrinogen (FI) deficiency	Cryoprecipitate, FFP
RiaSTAP (CSL Behring)	pd-fibrinogen	Yes	Fibrinogen (FI) deficiency	PCC, FFP
Fibryga (Octapharma)	rFVIIa	N/A	FVII deficiency	PCC, FFP
NovoSeven (NovoNordisk)	pd- FX	Yes	FX deficiency	FFP
Coagadex (Bio Products Laboratory)	pd-FXI	Yes	FXI deficiency	FFP, Cryoprecipitate
Hemoleven (LFB) (unlicensed) ^a	pd-FXIII	Yes	FXIII deficiency	FFP, Cryoprecipitate
Corifact (CSL Behring)	rFXIII-A	N/A	FXIII A-subunit deficiency	FFP
Tretten (NovoNordisk)	rFVIII or pd-FVIII	N/A (rFVIII)	FVIII deficiency (FV + FVIII)	
FVIII (numerous options)			FV (See below)	
PCCs	FII > FIX = FX > FVII	Yes	FII deficiency	FX deficiency, VKCFD
3-factor PCCs	FX > FII > FIX > FVII	Yes	FII deficiency	FX deficiency, VKCFD
Profilnine (Grifols)	FX > FII > FIX > FVII	Yes	VKCFD	FII and FX deficiencies
Bebulin (Shire)	FII > FX > FIX > FVII	Yes	FII deficiency	FX deficiency, VKCFD
4-factor PCCs	FX > FII > FIX > FVII	Yes	VKCFD	FII and FX deficiencies
Kcentra (CSL Behring)	All coagulation factors	No	FV deficiency	All RBDs
Octaplex (Octapharma) ^a	FVIII, fibrinogen, VWF, and FXIII	No	FXI deficiency	VKCFD
Beriplex (CSL Behring) ^a				Fibrinogen deficiency
FFP				FXIII deficiency
SD-Fresh Frozen Plasma				
(Octaplas, Octapharma)				
Cryoprecipitate				

^a Not available in USA.**Table 2**

Recommended Dosing for Treatment of Bleeding in RBDs.

Deficient factor and disease severity [27]	Plasma half-life	Recommended trough levels [17,24]	Incremental recovery, (Median) [24]	On-demand dosages [2,24]
Fibrinogen (FI) A-fibrinogenemia (fib) - absent Severe hypo-fib - < 0.5 g/L Hypo-fib-0.5–1.5 g/L Dys-fib- antigen:activity ratio < 0.7	2–4 days	Minor bleeding: 1 g/L Major bleeding: 1.5 g/L	0.017 g/L per mg/kg	pd-FI (50–100 mg/kg) Cryoprecipitate (15–20 ml/kg) SD-FFP (15–30 ml/kg)
Prothrombin (FII) Severe: undetectable Moderate: < 10% Mild: > 10%	3–4 days	Minor bleeding: > 10% Major bleeding: > 20%	PCC: 2 IU/dL per IU/kg FFP: ~2IU/dL per IU/kg	pd-PCC (20–40 IU/kg) SD-FFP (15–25 ml/kg)
FV Severe: undetectable Moderate: < 10% Mild: > 10%	36 h	Minor bleeding: 10% Major bleeding: > 15-20%	~ 1 IU/dL per IU/kg	SD-FFP (15–25 ml/kg)
FVII Severe: < 10% Moderate: 10-20% Mild: > 20%	4–6 h	> 20%	0.44 IU/dL per IU/kg	rFVIIa (15–30 mcg/kg) pd-PCC (20–30 IU/kg)
FX Severe: < 10% Moderate: 10–40% Mild: > 40%	40–60 h	Minor bleeding: 20% Major bleeding: > 40%	pd-FX: 2.08 IU/dL per IU/kg PCC: 1.7–2 IU/dL per IU/kg	pd-FX (20-30 IU/kg) pd-PCC (20–30 IU/kg) SD-FFP (10–20 ml/kg)
FXI Weak clinical-lab correlation	40–70 h	15-20%	pd-FXI: 2 IU/dL per IU/kg	SD-FFP (15-20 ml/kg) pd-FXI (15-20 IU/kg) ^a
FXIII Severe: undetectable Moderate: 30% Mild: > 30%	11–14 days	30%	pd-FXIII: 2 IU/dL per IU/kg [75] r-FXIII-A: 1.7 IU/dL per IU/kg	pd- FXIII (20-40 IU/kg) r-FXIII-A (35 IU/kg) Cryoprecipitate (2-3 bags) SD-FFP (3 ml/kg) r-FVIII (25–40 IU/kg) SD-FFP (15–25 ml/kg)
FV + FVIII Severe: < 20% Moderate: 10–40% Mild: > 40%	36 h (FV) 10–12 h (FVIII)	Major bleeding: FVIII > 40% FV > 15-20%		
VKCFD (FII, FVII, FIX, FX) Weak clinical-lab correlation	As for each factor	No data available		- Vitamin K (10 mg) IV or SC for minor bleeding - pd-PCC (20–30 IU/kg) with Vitamin K (5–20 mg) for severe bleeding or major surgery - SD-FFP (15–25 ml/kg)

Note: pd, plasma-derived; r, recombinant; PCC, prothrombin complex; concentrate; SD, solvent/detergent; FFP, fresh frozen plasma.

^a Not available in USA.

Table 3
Recommendations for long-term prophylaxis therapy for RBDs.

Deficient factor	Recommended trough levels	Long-term prophylaxis dosages	Frequency of Prophylaxis
Fibrinogen (FI)	0.5–1 g/L	pd-Fibrinogen: 30–100 mg/kg Cryoprecipitate: 1 unit 3 units	1–2 times/week 3 times/week Every 7–10 days
Prothrombin (FII)	20–30%	pd-PCC: 20–40 IU/kg	1 time/week
FV	10–20%	FFP: 20–30 ml/kg	2 times/week
FVII	10–15%	rVIIa: 20–40 mcg/kg	2–3 times/week
FX	10–20%	pd-FX: 20–40 IU/kg	2 times/week
FXI	15–20%	No data	–
FXIII	2–5%	pdFXIII: 10–40 IU/kg r-FVIII-A: 35 IU/kg Cryoprecipitate: 2 units	Every 4–6 weeks Every 4 weeks Every 3 weeks
FV + FVIII		Vitamin K: 5–20 mg	1 time/week orally
VKCFD (FII, FVII, FIX, FX)	No data	Vitamin K: 5–20 mg	1 time/week orally

Note: pd, plasma-derived; r, recombinant; PCC, prothrombin complex concentrate.

including large deletions, frame-shift, splice-site and early truncating nonsense mutations are the most frequently identified mutations, irrespective of geographic locations of affected individuals [68]. Mutations in the fibrinogen alpha gene account for the majority of cases of congenital afibrinogenemia [68]. Two common mutations, both in the FGA gene (a 11 kb deletion and a splice site c.510 + 1G > T mutation), should be screened first in all new kindreds before investigating FGB and FGG. A recent report on FGB mutations dictating the structure for fibrinogen assembly and secretion may provide some more insight into bleeding versus thrombotic phenotypes [69].

Currently, the databases that are actively collecting data on phenotype and genetic variants of RBDs are (1) RBDD project (<http://eu.rbdd.org>, accessed on March 7, 2018); (2) the International Society on Thrombosis and Haemostasis (ISTH) website (<http://www.isth.org/?MutationsRareBleeding>, accessed on March 7, 2018); and (3) The European Association for Haemophilia and Allied Disorders Coagulation Factor Variant Databases (<http://www.eahad-db.org/citations.html.php>, accessed on March 7, 2018). Recently, ISTH sponsored the ThromboGenomics project (<https://haemgen.haem.cam.ac.uk/> accessed on March 7, 2018) which will help to streamline the diagnosis of patients with RBDs by developing a comprehensive genetic database for and with the use of next generation sequencing. As mutation detection becomes easily available and less expensive, it would be important to know if novel polymorphisms detected are the cause of the specific clotting factor deficiency by obtaining data on allele frequency in the genetic population. The continuous characterization of novel molecular defects combined with functional studies will provide a better understanding of the complexity of these disorders and its implications on phenotype.

4. Management

The rarity of RBDs hampers the development of randomized controlled trials and there is consequently absence of evidence-based guidelines. The recommendations regarding the management of bleeding episodes, surgery, and pregnancy (including care of the potentially affected newborn) are mainly based on existing literature and expert consensus [24]. In general, the need for hemostatic therapy is based on severity of disease, type of bleeding episode, baseline residual factor activity and particularly, for surgical procedures (minor or major) it depends on site of bleeding (areas with abundant fibrinolysis), and personal or family history of bleeding with previous hemostatic challenges or a history of thrombosis.

4.1. Replacement products

Replacement products available in the United States (U.S.) for treatment or prevention of bleeding in RBDs, when indicated, are

summarized in Table 1. When available, replacement of the deficient factor with specific single-factor concentrates is the treatment of choice (Table 2). Single factor concentrates are available for the majority of deficient factors except for FII and FV and for FXI deficiency, a concentrate is available only in Europe. Most of the factor products are plasma-derived (PD) with recombinant options available for FVII and FXIII deficiency. Some of the RBDs (FI, FXI) may also be associated with thrombo-embolic events which needs to be taken into consideration while planning treatment.

In the absence of specific factor concentrate, alternate sources of factor replacement include Fresh Frozen Plasma (FFP) and cryoprecipitate. Prothrombin complex concentrates (PCCs) are highly purified concentrates of specific coagulation factors obtained from pooled normal plasma, namely FII, FIX, FX, and FVII in varying amounts depending on the product [70,71]. For FII deficiency, prothrombin complex concentrates (PCCs) are the preferred replacement product of choice over FFP as they are subjected to a process of viral inactivation [70]. FFP is the mainstay of treatment for FV deficiency, in the absence of any other products containing this factor. In patients with combined FV + FVIII deficiency, hemostasis for severe bleeds/major surgeries is achieved by administering FFP to replace FV along with FVIII concentrate to replace FVIII [72,73]. For minor bleeding episodes/surgeries, desmopressin (DDAVP) can also be used to increase FVIII levels [72–74]. In the U.S., FFP or solvent detergent treated FFP may also be used in FXI deficiency when hemostasis may be required during trauma or surgery [70]. FFP or PCCs are recommended for treatment of bleeding or for surgical coverage in VKCFD, in addition to maintenance oral vitamin K [42].

Cryoprecipitate, a byproduct of FFP obtained after thawing a single-donor volume of FFP at 4 °C, is rich in von Willebrand factor, FVIII, FXIII, and FI and is most often used for replacement of FI and FXIII, although more recently, purified plasma-derived and recombinant products have become available to treat these latter two deficiencies [70,75–77].

Although widely available and relatively inexpensive, FFP and cryoprecipitate have several disadvantages compared with single-factor concentrates and PCCs. They have variable factor levels which makes accurate dosing a challenge and often requires large volumes to be infused with the risk for fluid overload and TRALI (Transfusion Associated Acute Lung Injury). They are not subjected to a viral inactivation process and thus carry the potential risk of pathogen transmission. They also require blood group matching prior to transfusion and the thawing process which can delay the administration time.

4.2. Ancillary therapies

Ancillary therapies that may be used as adjuncts to factor replacement include systemic antifibrinolytics (ie, tranexamic acid and ε-

Table 4
Clinical characteristics and management of inhibitors in FV, FVII, FXI and FXIII deficiencies.

Alloantibody	FV deficiency [91–94]	FVII deficiency [95,96]	FXI deficiency [97–102]	FXIII deficiency [103]
Type	Low and high titers	Low and high titers	Low and high titers	High titer and transient non-neutralizing low titer
When to suspect	Recurrent bleeding, Anamnestic response or failure of correction of PT/ aPTT following FFP use	Recurrent bleeding, Anamnestic response following rVIIa	Recurrent bleeding, failure of correction of prolonged aPTT following FFP/ FXI concentrate in patients with null mutations rVIIa, FXI concentrate in higher doses; TXA pre-op	Recurrent, severe bleeding, anamnestic response following FXIII concentrate
Treatment of bleed	First line: platelets, rVIIa, FEIBA Second line: high dose FFP, plasmapheresis	rVIIa		FXIII concentrate
Inhibitor eradication	Steroids, CTX, Rituximab	Steroids, CTX, azathioprine, immunoglobulins	Steroids, CTX	CTX, IVIG, high dose FXIII concentrate
# of patients reported	6	13	19	4

Note: CTX: cyclophosphamide; IVIG: intravenous immunoglobulin; TXA: tranexamic acid.

aminocaproic acid), topical hemostatic agents (eg, fibrin glue, collagen). Antifibrinolytics may be used alone for less severe forms of mucosal tract bleeding, minor (eg, dental) procedures or heavy menstrual bleeding, however caution needs to be exercised when using to treat hematuria. In addition, systemic and local hormonal therapies are often effective and considered first line therapies for management of heavy menstrual bleeding [45,78].

4.3. Prophylactic replacement therapy

Prophylactic hemostatic therapy in RBDs, primary or secondary, may be considered in relation to bleeding frequency, risk of severe spontaneous bleeding, and associated long-term disability (Table 3). Primary prophylaxis has been effectively used in boys with severe HA to reduce joint damage from recurrent hemarthroses [79]. However, little is known regarding the use of prophylaxis in RBDs. The only RBD for which prophylaxis is routinely recommended is severe FXIII deficiency, in which prophylaxis is recommended from the time of diagnosis because of the risk for serious, even life-threatening, bleeding events such as ICH even later in life [80]. Secondary prophylaxis may be considered after musculoskeletal bleeding or life-threatening hemorrhage, particularly in deficiencies of FXIII, FX, and FVII and in severe cases of FI and FV deficiencies [4,81,82]. The risk of developing thrombosis with FI deficiency (even with afibrinogenemia) should be considered and treatment should be tailored to the bleeding-thrombosis phenotype in the individual and family. Prophylactic fibrinogen replacement to maintain trough levels > 1 g/L during pregnancy and > 1.5 g/L at labor is also required for those with severe deficiency or with previous adverse outcomes [24]. Parenteral administration of high-dose VK is also effective in complete or partial correction of the coagulation factor defects associated with inherited VKCFD.

4.4. Novel therapeutic strategies

Recent development of non-replacement therapies that promote hemostasis by different mechanisms may have a promising role in the treatment of patients affected with RBDs. One such novel hemostatic drug is a bispecific recombinant antibody, Hemlibra (emicizumab-kxwh), which was recently approved by FDA (November 2017) for use in hemophilia A patients with inhibitors which leads to effective thrombin generation by binding to FIXa and FX, thus mimicking the cofactor activity of FVIII [83,84]. Other approaches to increase thrombin generation which may be revolutionary in the treatment of RBDs include downregulation or inhibition of activity of natural anticoagulants such as anti-tissue factor pathway inhibitor (TFPI) monoclonal antibody (concizumab) and RNA interference molecule (ALN-AT3/Fitusiran) targeting antithrombin III [85–88]. Other targets that have been explored are activated protein C and an engineered factor Va mutant which have shown improved thrombin generation and hemostasis in hemophilia mice models [89,90].

These alternative therapeutic agents have demonstrated favorable safety profile after either weekly or monthly subcutaneous injections and have been effective in hemophilia. Their role needs to be explored in RBDs especially with FVII/FXI/FX deficiency but might not be useful for other RBDs such as dysfibrinogenemia or FXIII deficiency.

5. Conclusions

Development of guidelines for management of spontaneous, trauma, post-surgical, delivery or post-partum bleeds for each type of RBD would require a detailed evaluation of each factor deficiency so that the hemostatic level of each single factor can be determined. This can be achieved by developing accurate data collection tools through international registries using common definitions, standardizing factor assay methodology, evaluating thrombin generation through TG assays and studying other disease modifiers that might be contributing to the

phenotypic heterogeneity of these disorders. Screening for inhibitors in RBDs with anamnestic response to replacement products warrants further study. Search for alternative indices or other potential modifiers of bleeding severity related to RBDs merits further research. Finally, conducting in vitro expression studies of each gene mutation is important to establish the mechanistic link between the gene mutations and the observed disease phenotypes.

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